

Version-Aware Normalization of Findings Against MITRE ATT&CK v19

Typed Data Models and Revocation-Resolving Technique Lookup for Security Tooling

CONTRIBUTION

A typed ATT&CK v19 data layer that normalizes findings across version churn by resolving revoked/renamed technique IDs through a maintained remap table over SHA-256-verified STIX data, with indexed lookup for downstream security tooling.

METHOD

- 01

Download/validate ATT&CK STIX data with SHA-256 verification.
- 02

Parse techniques/tactics/platform metadata into typed structures.
- 03

Resolve deprecated/revoked IDs through the v19 remap table.
- 04

Build in-memory indexes for ID, tactic, platform, and keyword lookup.
- 05

Regression-test migration and data-loader behavior across supported Python versions.

BOUNDARIES / LIMITATIONS

- ATT&CK mapping/normalization does not prove an attack occurred.
- O(1) dictionary/index lookup describes the in-memory data structure, not end-to-end system latency.
- Future ATT&CK releases require new migration/remap evidence.

CURRENT VERIFIED EVIDENCE

165	Tests passed
63.83%	Statement coverage
113	Test functions
11	Test files
22	Revoked technique IDs

EVIDENCE NOTES

29	Total remaps
165 tests passed.	
48	New techniques in documented v19 migration
63.83% statement coverage; CI gate is 60%.	
113 test functions across 11 test files; parametrization expands these to 165 executed tests.	

REPRODUCE

```
git clone https://github.com/poojakira/attack-v19-core.git
cd attack-v19-core
git checkout 22cc9ee3e08406a446bf500f84ea1d8fb53a6a6d
python -m pip install -e ".[dev]"
pytest tests/ -q --cov=attack_v19_core --cov-report=term
```